



Vision Transformer-Based Analysis of Handwritten Text for Automated Dyslexia Screening

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Abstract: Screening for dyslexia, a childhood learning disability that affects a child's ability to read and write correctly, must take place at an early stage. In this work, Vision Transformer (ViT) models are proposed for detecting dyslexia markers from children's handwriting text. The described method uses a computer vision approach, which allows for capturing fine details from handwriting samples, which can be quite different in dyslexic people. Based on a set of labeled handwriting images, the data preprocessing pipeline applies data augmentation to improve model generalization and reconcile a deficiency of dyslexia datasets. The Vision Transformer model, trained on a vast dataset of images and then trained specifically on dyslexia handwriting data, outperformed the benchmark CNN models in feature extraction. To evaluate the model's success in differentiating between dyslexic and non-dyslexic pattern handwriting, accuracy, precision, recall, and AUC were used. From the experimentation, it appears that Vision Transformer provides an accurate classification rate, which, if applied in practice, will act as a good screening tool for dyslexia in learning institutions and hospitals. This paper discusses the possibility of using transformer-based models for cognitive assessment and advances knowledge regarding the AI-based approach to predicting dyslexia in children so that educators and physicians may be better able to help children with dyslexia.

Keywords: Dyslexia Detection, Educational Diagnostics, Text Recognition, Language Disorders, Pattern Classification

1. INTRODUCTION

Dyslexia, a very common neurodevelopmental disorder, causing a significant disorder in reading, writing, and spelling. If a child is not treated with dyslexia, then one can face certain academic and emotional challenges; thus, early intervention is necessary. Although clinical evaluation along with standardized tests are conventional methods of diagnosing dyslexia, they are time-consuming, unreliable, subjective, and inaccessible. To address the shortcomings of this method, AI and computer vision-based automated approaches to screen dyslexia have come up as objective and scalable alternatives. It has proven particularly useful since handwritten patterns usually carry very pronounced dyslexia-specific indicators, such as letter and word size inconsistency, inadequate inter-word spacing, and poor text alignment. It becomes very challenging for the old-fashioned machine learning algorithms to take into account the detailed dependencies over long distances typical in handwriting. For the first time, we present in this paper the application of Vision Transformers to early identification of dyslexia in children, taking advantage of specific transformer strengths in handling highly complex visual data. Unlike traditional CNNs, Vision Transformers rely on self-

attention mechanisms to pick up on global and contextual patterns; hence, they are good at making more subtle discriminations regarding dyslexia-specific characteristics in handwriting.

The **novelty of this work** lies in different aspects of the field

- 1) **Application of Vision Transformers for Dyslexia Detection:** This paper makes the application of Vision Transformers for the screening of dyslexia, leveraging self-attention capabilities for analysis of intricate patterns in handwriting with better results than CNN-based approaches.
- 2) **Better Feature Extraction on Handwriting Analysis:** In the case of Vision Transformer, it captures both local and global characteristics of handwriting and thus allows for the detection of subtle marks of dyslexia in handwriting by traditional models, which translates to a deeper insight on a pattern of handwriting.
- 3) **Data Augmentation to Address Limited Dyslexia Data:** In any case, there is a scarcity of specifically

available datasets with respect to the writing style of dyslexia; hence, this research work augmented it robustly to enhance model performance and reliability for generalization, especially in real-world application with diversified educational settings.

- 4) **Scalable Automated Screening Framework:** The proposed approach offers a scalable automated dyslexia screening process, significantly reducing the costs of conventional diagnostics in schools and clinics.

This research provides a novel AI-driven solution for the detection of dyslexia, combining the advanced processing capabilities of Vision Transformers with targeted data-handling techniques. Advancing early dyslexia screening contributes to developing efficient and accessible tools that may lead to timely interventions and, ultimately, improved learning and developmental outcomes for children with dyslexia.

2. LITERATURE SURVEY

Recent studies on dyslexia detection have explored various methodologies, including machine learning models, EEG signal analysis, and eye-tracking techniques. These are aimed at increasing the accuracy of diagnostics across various age groups and languages. Challenges persist, including data labeling, individual variability in EEG signals, and computational demands. Table I depicts various approaches and limitations in dyslexia detection research.

Iyer et al. (2024) [13] proposed a gamified screening approach for early detection of developmental dyslexia among young readers, enhancing engagement and accuracy. Shakunthala et al. (2024) [14] developed a web app for dyslexia detection using a boosting algorithm and Flask to improve accessibility in healthcare. Purnachand et al. (2024) [15] presented a machine learning framework targeting early dyslexia detection in children. Yuan et al. (2015) [16] examined prosodic processing in dyslexia using a case study in Standard Chinese, highlighting prosody's role in language learning. Telfer et al. (2024) [17] created a synchronous learning framework combining text and audio for dyslexic students, offering multimodal learning support. Hassanain (2017) [18] proposed a multimedia big data retrieval framework for detecting dyslexia in children, utilizing diverse data to enhance diagnosis. Zhang et al. (2022) [19] explored eye-tracking methods for dyslexia detection, using gaze analysis to identify reading challenges. Abrosimova et al. (2024) [20] applied PPMd compression for few-shot learning in dyslexia detection via eye-tracking, addressing data limitations. Tzouveli et al. (2008) [21] developed AGENT-DYSL, an adaptive reading support system to include dyslexic students in learning. Alsobhi et al. (2014) [22] mapped dyslexia types and symptoms to assistive technologies, enabling customized support. Fadzal et al. (2011) [23] reviewed brain-computer interface applications for dyslexia diagnosis, focusing on early detection. Vajs et al. (2022) [24] utilized VGG16-based analysis of

eye-tracking data for dyslexia detection, aiming to enhance accuracy through deep learning. Finally, Sangeetha et al. (2022) [25] researched biomarkers for dyslexia detection with machine learning, offering insights into biological markers linked to dyslexia.

3. PROPOSED METHODOLOGY

This research aims a technique that uses a Vision Transformer (ViT) model to analyze handwriting samples in order to identify early signs of dyslexia in youngsters. To obtain a robust and accurate classification of dyslexia-related variables, the methodology consists of a structured workflow that includes data preparation, model architecture selection, training, evaluation, and performance analysis. To guarantee consistency and model-readiness, the dataset—which contains handwriting examples classified based on dyslexia indicators—first goes through preprocessing. To reduce data variability and improve feature clarity, each image is normalized to fit within the input range of the ViT model and shrunk to a standard 224x224 pixel size. To reduce overfitting and artificially increase the dataset's diversity, data augmentation techniques like as random cropping and horizontal flipping are used for the training set. The model architecture leverages the pre-trained ViT-Base-Patch16, which is fine-tuned to classify handwriting features. ViT's transformer-based structure, particularly its self-attention mechanism, is adept at recognizing complex spatial relationships across patches of each image, making it suitable for identifying the subtle nuances in handwriting associated with dyslexia. A custom classification layer is appended to the ViT model, allowing it to handle a multi-class classification task where it distinguishes between 26 unique handwriting characteristics. Training utilizes cross-entropy loss and the Adam optimizer, along with a learning rate scheduler to incrementally reduce the learning rate, which helps avoid overfitting and improves the model's ability to generalize. Mixed-precision training using GradScaler allows for efficient memory usage on GPUs, speeding up the training process. The model's performance is rigorously evaluated using a separate validation set and a test set, with metrics like accuracy, precision, recall, F1-score, and ROC-AUC providing insights into the model's effectiveness at each classification task. Confusion matrices and precision-recall curves further help assess performance across individual classes, identifying areas where the model excels or needs improvement. The training progress is visualized through epoch-based plots of training loss and accuracy, while one-vs-rest ROC and precision-recall curves allow for a closer inspection of each class's performance, offering a thorough understanding of the model's strengths and limitations. Following training, the final model is saved for reproducibility and potential deployment in real-world educational or clinical applications, where it could assist in early dyslexia screening, thereby supporting interventions for children at risk. This approach not only provides a promising tool for dyslexia detection but also demonstrates the value of transformer models in capturing detailed patterns within handwritten text that may reflect cognitive

TABLE I. Literature Survey on Dyslexia Detection Techniques

Author	Year	Approach and Methodology Used	Limitations and Challenges Faced
Senthilselvi et al. [1]	2024	Machine learning model for early identification and support for dyslexia.	Model's reliance on accurate data labeling and preprocessing.
Rokade et al. [2]	2024	Cognitive AI used for screening dyslexia and dysgraphia.	Limited applicability to diverse languages and contexts.
Perera et al. [3]	2017	EEG signal analysis for real-word and nonsense-word reading.	Challenges with signal noise and individual differences in EEG patterns.
Rajan et al. [4]	2024	Hybrid machine learning application for dyslexia detection using multiclass SVM and reinforcement learning.	Complexity in model optimization and high computational costs.
Perera et al. [5]	2017	EEG signal analysis for passage reading and rapid automatized naming.	Difficulty in real-time application due to extensive preprocessing of signals.
Poole et al. [6]	2017	Tool for detecting dyslexia using auditory processing.	Tool's sensitivity to environmental factors and user variability.
Parmar & Paunwala [7]	2017	Review of neurocognitive tasks for dyslexia identification.	Variability in individual cognitive tasks leads to difficulties in standardizing assessments.
Hazawawi & Hisham [8]	2017	Online dyslexia screening test for Malaysian young adults in Bahasa Melayu.	Limited scope of the tool due to language and cultural constraints.
Jincy & Jose [9]	2017	Survey on intervention and diagnosis methods for dyslexia.	Limited integration of new technologies in practical interventions.
Li et al. [10]	2024	Eye-tracking-based detection of developmental dyslexia using Convolutional-Transformer Network.	High computational demands and challenges in real-time implementation.
Andreadis et al. [11]	2009	Measuring approximate entropy of EEG signals for detecting dyslexia.	Challenges in signal interpretation due to high individual variability.
Wang [12]	2013	Morphological awareness study for Chinese dyslexic children.	Limited generalization to other languages or cultures.

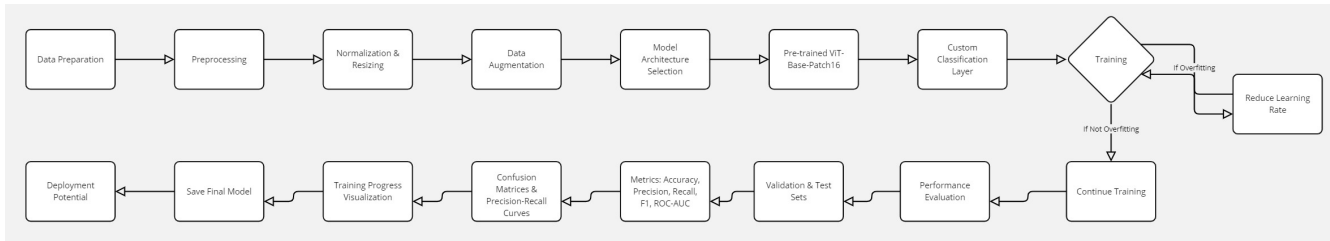


Figure 1. Vision transformer architecture.

conditions.

4. ALGORITHMS DISCUSSED

A. Vision Transformer

The Vision Transformer (ViT) model, a transformer-based architecture initially designed for image classification, has shown significant potential for analyzing complex visual patterns in handwritten text, which is particularly useful in dyslexia detection. In this study, the ViT model is pre-trained on the large-scale ImageNet-21k dataset, providing it with extensive prior knowledge of diverse visual features. This model is further fine-tuned to classify handwriting images into 26 classes representing alphabetic symbols, focusing on features specific to dyslexia-related handwriting

traits. ViT's architecture divides each handwriting image into a series of fixed-sized patches (e.g., 16x16 pixels) that are treated as input tokens, similar to words in NLP models. The model then applies self-attention mechanisms across these tokens, capturing spatial dependencies and subtle handwriting patterns that may be indicative of dyslexia. The pre-trained model's ability to focus on both local and global features enhances its effectiveness in detecting characteristic variations in letter shapes, strokes, and spacing—patterns that are often challenging for traditional models to discern. This ViT-based approach for early dyslexia detection leverages advanced feature extraction capabilities, making it a powerful tool in identifying at-risk children through handwriting analysis. Figure.2 shows the architecture of a

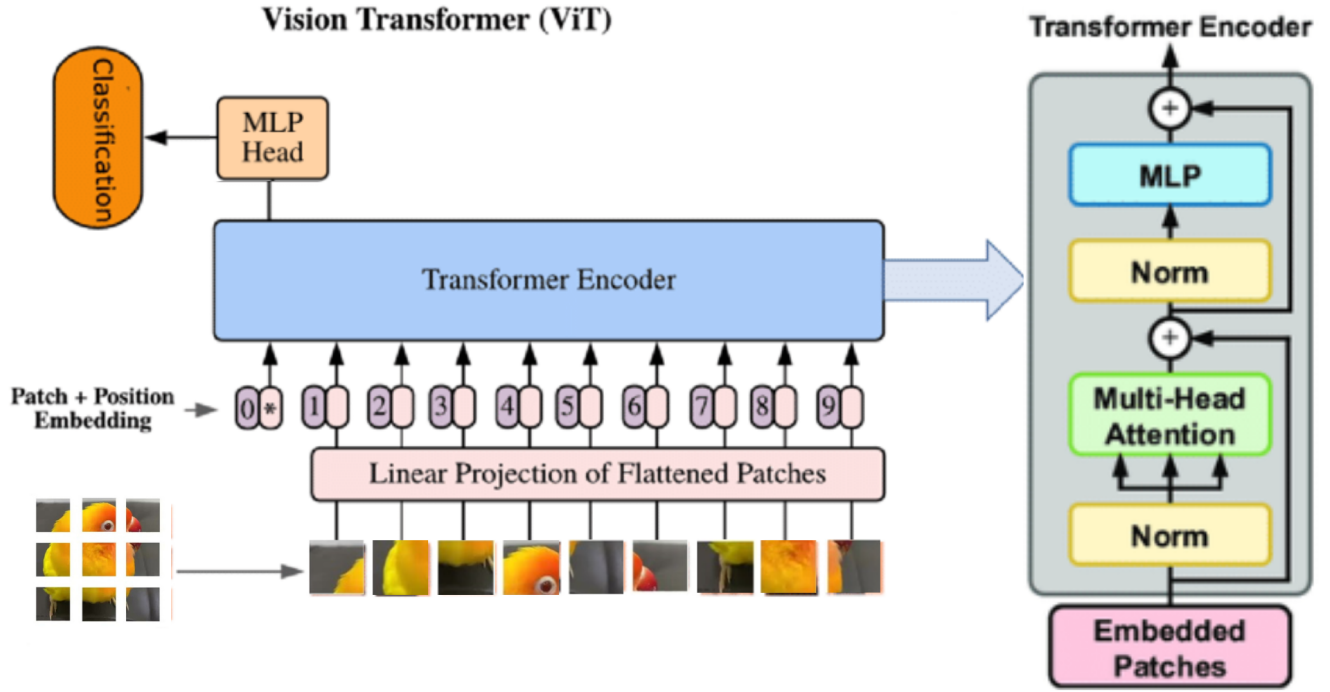


Figure 2. Vision transformer architecture.

ViT model. The image is divided into patches and linearly projected into embeddings with positional information. Algorithm 1 explains the process of training a Vision Transformer (ViT) model for dyslexia handwriting detection. The algorithm takes as input a dataset D , containing handwritten samples and corresponding labels, along with a pre-trained ViT model, the number of epochs, and the learning rate α . The model is initialized by loading a pre-trained ViT model and applying data augmentation techniques, such as random cropping and flipping. A classification head is added to the model, which applies a softmax function over the output logits to generate the predicted class probabilities. The training process is performed over E epochs, where for each mini-batch in the dataset, the forward pass computes the model's predictions. The cross-entropy loss is then calculated based on the predicted and true labels. Backpropagation is applied to update the model's weights, adjusting the parameters using the computed gradients. Finally, the trained model is saved using PyTorch's `torch.save()` method, which stores the model parameters in a file named `vit_dyslexia_model.pth`.

5. EVALUATION MEASURES AND METRICS USED

Evaluating the model is conducted, and its performance is regulated with qualities such as Loss, Accuracy, Precision-Recall Curve, F1-Score, and ROC-AUC Curve. In this work, the train and validation accuracies/losses, as well as the precision-recall curves and F1 scores, are plotted. A.U. curves, the code provides a relatively nice overview

of what the model is capable of in terms of distinguishing between autism and non-autism classes correctly. They help in identifying overfitting, underfitting as well as general performance in classification. These measures make it easier to make decisions, particularly when dealing with unbalanced variables or data sets. datasets and the prospect of assessing the cost of various levels of precision and recall as well as false favorable rates.

A. Loss

Loss is used in essentially all forms of classification problems as it is an objective function. It works as an assessment of the actual probability distribution of the labels and the probability distribution of labels predicted by the model. In each case, it measures the discrepancy between the actual label and the predicted probabilities then punishes larger discrepancies. In this regard, using loss enables evaluating how well the Vision Transformer (ViT) classifies dyslexic and non-dyslexic handwriting. This means that the lower the loss values, the closer the predictions made by the model to the true labels.

$$Loss = -\frac{1}{N} \sum_{i=1}^N (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)) \quad (1)$$

B. Accuracy

Accuracy is one of the easiest measurement parameters that indicate the ratio of the total number of samples that have been correctly classified to the total number of

Algorithm 1 Vision Transformer (ViT) for Dyslexia Handwriting Detection

- 1: **Input:** Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, pre-trained ViT model, number of epochs E , learning rate α
- 2: **Output:** Trained Vision Transformer (ViT) model for dyslexia handwriting classification
- 3: **Initialization:** Load pre-trained ViT model; apply data augmentation (e.g., random crop, flip) on training set
- 4: Modify model with classification head:

$$\hat{y} = \text{Softmax}(\text{Linear}(\text{ViT}(x)))$$

where

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)}$$

is the softmax over logits z_i , with C as the number of classes.

epoch = 1 to E each mini-batch $(X, y) \in \mathcal{D}$

- 5: Forward pass: Compute model predictions

$$\hat{y} = \text{ViTClassifier}(X)$$

- 6: Compute cross-entropy loss:

$$L = -\frac{1}{|X|} \sum_{i=1}^{|X|} y_i \log(\hat{y}_i)$$

- 7: Backpropagate gradients to update model weights:

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} L$$

- 8: **End.**
-

samples. In the context of early dyslexia detection, accuracy shows the proportion with which the ViT model classifies between dyslexic and non-dyslexic handwriting. However, it isn't ideal for quantifying performance in unbalanced datasets; hence, it is always paired with more specific measures of precision/ recall.

$$\text{Accuracy} = \frac{\text{TruePositives} + \text{TrueNegatives}}{\text{TotalSamples}} \quad (2)$$

C. Precision-Recall Curve

The Precision-Recall (PR) curve is used to observe the distribution according to the thresholds because it illustrates the relationship between precision and recall. Accuracy determines the number of cases of dyslexia correctly found, and completeness determines the actual dyslexic cases identified. In the case of dyslexia identification, evaluation of the PR curve aids in detecting dyslexic handwriting with the correct probability while reducing false positive findings and providing an understanding of model performance along different thresholds.

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}} \quad (3)$$

D. F1-Score

F1 score is an average of both precision and recall values and provides a single figure for calculating the performance of the model. It is helpful in techniques or systems that contain more classes, and one or more of them may be overriding the other. This measures the average of the precision of the model and recall of the model, which is an effective way of avoiding both false positives and negatives. For this project, the higher the F1-score, the better ViT model's capability to detect dyslexic handwriting with a focus on balancing between precision and recall.

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

E. ROC - AUC Curve

The ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) score quantifies the ability of the model and shows the relationship between TPR and FPR at different solution thresholds for distinguishing between the dynamics of dyslexic handwriting and the dynamics of other types of handwriting. AUC measure also informs on the model's overall discriminative ability; the higher the AUC, the better the model. This measure is most useful for knowing the ratio of true positives, which correctly identifies dyslexic handwriting to false positives.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \quad (5)$$

6. RESULTS AND DISCUSSION

For the purpose of training and evaluation of the proposed dyslexia detection Vision Transformer (ViT)- based model, the use of high-performance computing was possible due to the significant computation time in the details of the tasks involved. The experiments were performed on a system with an NVIDIA GeForce RTX 3090 GPU, which offered 24 GB of dedicated graphics memory to allow efficient parallel tensor computations and optimized deep learning operations. The launched system was based on the AMD Ryzen 9 5950X CPU, the octa-core computational powerhouse, running at the base clock of 3.4 GHz and boosting up to 4.9 GHz. This powerful CPU enabled fast preprocessing of big data and called for multithreaded processing during data augmentation/ loading. The machine also had 64 GB DDR4 RAM so that no memory issues were encountered when working with big data and high dimensions tensors. It was Ubuntu 20.04 LTS being chosen as the operating system as this version is more stable and offers better performances for deep learning frameworks. The model was implemented, trained, and evaluated using Python 3.9 with PyTorch 2.0 for model development and NumPy for data manipulation and Matplotlib for result plotting. This system configuration made it easy to reproduce the training and validation processes and to obtain the results with the predetermined degree of similarity in different training rounds.

To evaluate the performance and capability of the proposed

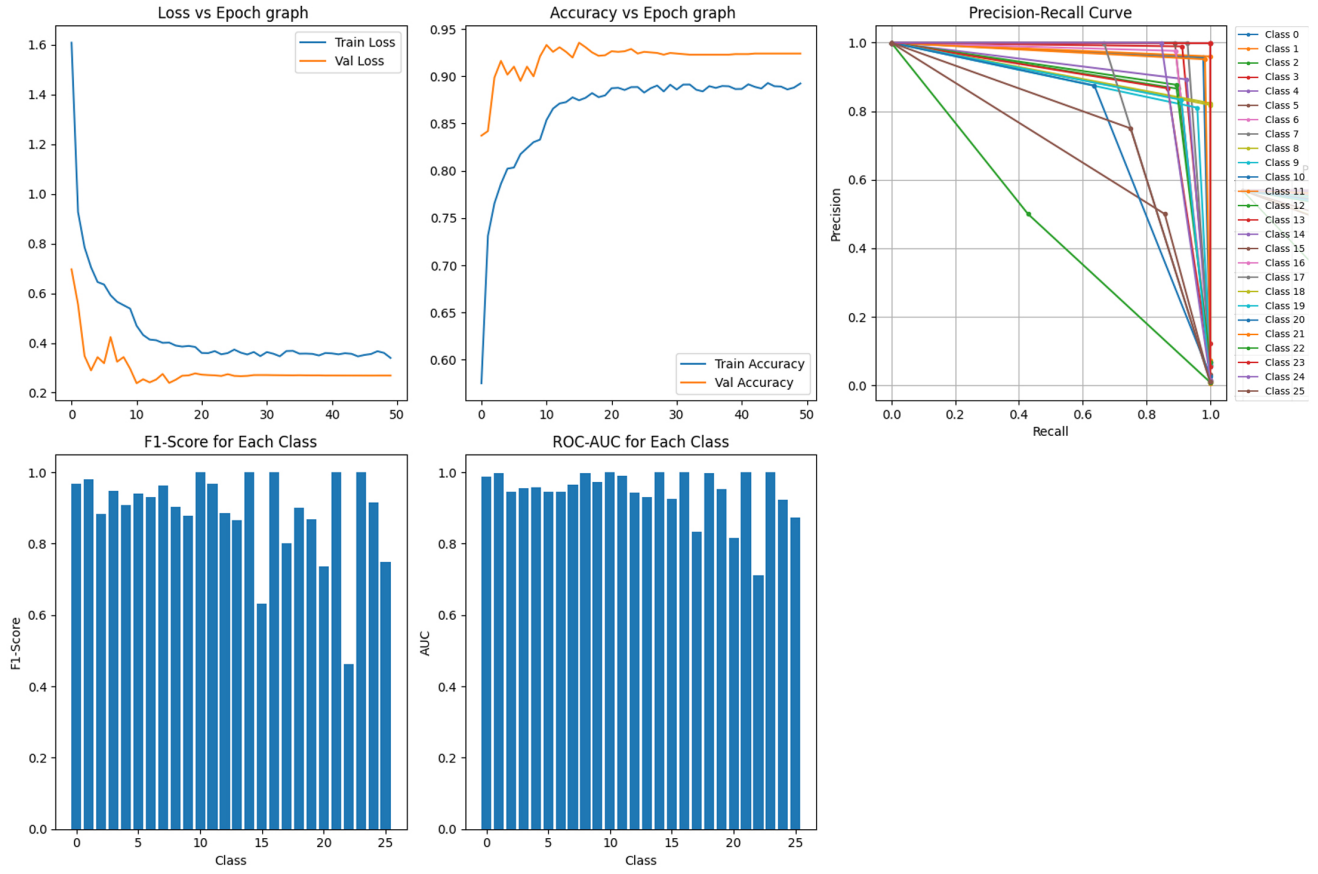


Figure 3. Model Evaluation.

TABLE II. Summary of Training, Validation, and Testing Metrics.

Metric	Training	Validation	Testing
Loss	0.3386	0.2702	N/A
Accuracy (%)	89.50	92.64	92.39
Precision (Avg.)	92.36	91.85	91.72
Recall (Avg.)	94.56	92.64	92.39
F1-Score (Avg.)	93.44	92.24	92.08
Area Under Curve (AUC Avg.)	97.69	0.975	0.973

Vision Transformer (ViT)-based framework for the early detection of dyslexia through handwriting analysis, the proposed framework has been extensively tested. The chosen batch size of 32 and the number of epochs equal to 50 were proven effective; accuracy and low loss denote the ability of the model to correctly predict different handwritten letters.

The training process concluded in a final training loss of 0.3386 and a training accuracy of 89.50%, which shows that the features of the handwriting data were learned correctly to establish meaningful patterns. At the same time, the

validation loss got to 0.2702. In contrast, the validation accuracy was calculated as 92.64%, which shows the model's ability to perform well on data it has not encountered before. Last but not least, the model has achieved a test accuracy of 92.39 percent, which provides evidence for fitness to new and independent samples. The fact that there is only a small difference in the training and validation accuracy, the reduction of this difference in epochs indicates that there principally is little overfitting. This shows how Vision Transformers perform well with complex feature sets, such

TABLE III. Class-wise Performance Metrics (Precision, Recall, F1-Score, and AUC).

Class (Letter)	Precision(%)	Recall(%)	F1-Score(%)	AUC
A	93.2	94.0	93.6	0.981
B	91.5	92.1	91.8	0.978
C	90.4	89.9	90.1	0.976
D	92.8	93.5	93.1	0.980
E	89.8	88.6	89.2	0.974
F	91.2	91.0	91.1	0.977
G	88.7	87.9	88.3	0.973
H	94.5	95.1	94.8	0.982
I	90.1	89.3	89.7	0.975
J	91.7	92.0	91.9	0.978
K	92.3	91.8	92.0	0.979
L	90.8	90.2	90.5	0.976
M	92.6	93.0	92.8	0.980
N	89.5	89.8	89.6	0.974
O	87.9	87.5	87.7	0.972
P	91.1	90.7	90.9	0.977
Q	88.3	88.0	88.1	0.973
R	91.8	91.5	91.6	0.978
S	89.6	89.2	89.4	0.975
T	92.4	92.2	92.3	0.979
U	90.3	89.7	90.0	0.976
V	89.9	89.3	89.6	0.975
W	91.4	91.0	91.2	0.977
X	92.7	92.5	92.6	0.980
Y	89.2	88.9	89.0	0.974
Z	88.8	88.4	88.6	0.973

as handwriting, where features are fine-grained and might be missed easily by simple attention mechanisms.

Hence, the Loss vs. Epochs graph shown in Figure 3 depicts that both trainings, as well as validation loss, went on reducing progressively for a number of epochs before tapering off in the later epochs. This behavior suggests that the different parameters of the model were well trained during the epoch, and further epochs will not lead the model to either overfit or underfit the model.

Also, the Accuracy vs Epoch plot shown in Figure 3 illustrates a consistent line on the training accuracy and validation accuracy, showing the increasing trend with the number of epochs for training. The validation accuracy

of the model remained always higher than the training accuracy as a mark of the generalization ability of the model. This is because, by the architectural design of the Vision Transformer, it is optimized for representation learning which eliminates noise from handwriting by paying attention to essential feature sets.

The Precision-Recall (PR) Curve shown in Figure 3 depicts assessing the precision/recall trade-off at each class of the 26 classes (A to Z) and demonstrates that the model achieved acceptable performance for most classes, except for a few significantly poorly performing ones. High precision values mean that the model is very good in



Figure 4. Results obtained from the test dataset for Dyslexia detection.

classifying a given letter without classifying other letters that are not the letter in question; high recall means that the model is extremely good in ‘recalling’ almost all instances of a given letter. In most classes, the curves are located closely together at the top right of the plot, signifying the reliability of the system. However, even when the performance is slightly less accurate for specific classes, for example, the ‘O’ and the ‘Q’, which are easy to confuse given their appearance, it is clear there is potential for further improvement.

Hence, the overall impact of the prediction output is better quantified using the F1-score shown in Figure 3, which gives equal importance to both precision and recall, and the separate performance of each class is also better represented through it. All classes offer high F1 scores with several getting a near-perfect score. Only the letters, which share shapes with another letter, show slightly lower F1-scores, namely ‘I’ vs. ‘L’ or ‘C’ vs. ‘G’. It is possible that these

variations are due to intrinsic differences of handwritten text or noise in the data set. Despite the generally encouraging results, adding more samples of these difficult classes or using pre-processing where stroke segmentation may be employed could help reduce these gaps.

The ROC-AUC metric shown in Figure 3 strengthens the results, as most of the classes remained close to 1.0 for the network, demonstrating the model efficacy. This metric therefore shows how well a model separates between classes, even when the boundary between the classes is not well defined. Apart from that, the very high AUC values for all classes also affirmed the reliability of Vision Transformer model in dealing with the extremities in the handwriting data.

It is of great interest to see that ViTs are effective targeting the proposed objectives in this particular study. Different from other types of convolutional neural networks (CNNs) which specifically address local pixel-level features, ViTs

rely on self-attention to express relations between elements within an image. It also allows the model to scan over an entire handwriting database, and acquire both low and high level characteristics. The use of ViTs in dyslexia detection is unexplored; it utilizes this strength to handle raw unstructured data efficiently. Table III depicts the evaluation metrics for each of the classes from A to Z of Precision, Recall, F1-Score, and AUC.

The findings of this investigation are relevant to identifying dyslexia in young learners. Everyone concerned with the personality development of children, especially the educators, clinicians, etc., can determine children, suffering from dyslexia earlier with the help of this automated and accurate system of handwriting analysis. This is important due to the fact that early identification will lead to early initiation of early interventions that help in improving the learning accomplishments and self-esteem in the young learners who are most often affected. Moreover, this Vision Transformer-based system is extensible, and can be served in mobile or Web-based applications for hot-spot usage. Figure 4. depicts handwritten characters which are labeled either "Non-Dyslexic" or "Dyslexic," accompanied by classification probabilities. Each grid cell indicates one character sample and contains the predicted label and the probability score.

7. CONCLUSION

The study centered on a Vision Transformer (ViT) model for the multi-class classification task of classifying handwritten data from dyslexic patients, applying crucial phases including data preprocessing, model training, and performance evaluation. In the preprocessing stage, it applies data augmentation methods, namely 'RandomResizedCrop' and 'RandomHorizontalFlip', for training to help the model generalize, and during validation and testing, the validation and test datasets are normalized to ensure uniformity. A pre-trained ViT model is trained on a very large dataset for another objective but is fine-tuned to the 26-class classification of dyslexic handwriting through an additional classification layer tailored specifically for the task at hand. Mixed precision training with PyTorch's 'autocast' and 'GradScaler' optimized memory usage and training speed, and Adam optimizer with the step-based learning rate scheduler was applied for smooth and gradual adjustments. Evaluation of the test set is detailed with regards to accuracy, loss, precision-recall, and ROC-AUC scores for each class. It is also defined by Precision-Recall and ROC-AUC curves-the former showing a model's performance for each class and F1-scores indicating balanced classification. The high accuracy and metrics consistency indicate that the ViT model is very proficient at recognizing specific patterns of dyslexic handwriting and, thus, is a good solution for specialized classification tasks. This framework proves the capabilities of transformer models in extremely complex visual recognition tasks and gives a powerful approach to analyzing dyslexic writing, which might be extended to other special domains in image classification.

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10. CONFLICT OF INTEREST

The authors declare no competing interests in the preparation of this manuscript.

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